Job Leon Feldbrugge

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Referees

Prof. Neil Turok, University of Edinburgh, Edinburgh, United Kingdom and Perimeter Institute for Theoretical Physics, Waterloo, Canada (neil.turok@ed.ac.uk)
Prof. Ue-Li Pen, Academia Sinica, Taipei, Taiwan and Canadian Institute for Theoretical Astrophysics, Toronto, Canada (pen@cita.utoronto.ca)
Prof. Celine Boehm, University of Sydney, Sydney, Australia (celine.boehm@sydney.edu.au)
Prof. Rien van de Weygaert, University of Groningen, Groningen, The Netherlands (weygaert@astro.rug.nl)

Academic positions

2026 - present **Member** at the Institute for Advanced Study, Princeton, United States of America
2026 - present **Postdoctoral Fellow** at the Leung Center for Cosmology and Particle Astrophysics, National Taiwan University, Taipei, Taiwan
2021 - 2026 **Higgs Fellow** at the Higgs Centre for Theoretical Physics at the University of Edinburgh, Edinburgh, United Kingdom
2019 - 2021 **Postdoctoral fellow** at the Perimeter Institute, Waterloo, Canada
2019 - 2021 **Postdoctoral fellow** at Carnegie Mellon University, Pittsburgh, United States of America

Education

2015 - 2019 **PhD** Physics, Perimeter Institute, University of Waterloo
Advisor: Neil Turok
Thesis: Path integrals in the sky: Classical and quantum problems with minimal assumptions
Defense date: October 17, 2019

2014 - 2015 **Master** Part III Mathematics (*with distinction*), University of Cambridge
Committee: Paul Shellard and Tommaso Giannantonio
Thesis: Primordial non-Gaussianity and large-scale structure

2012 - 2014 **Master** Physics (*cum laude*), van Swinderen Institute, University of Groningen
2012 - 2014 **Master** Astronomy (*cum laude*), Kapteyn Institute, University of Groningen
2012 - 2014 **Master** Mathematics (*cum laude*), Bernoulli Institute, University of Groningen
Committee: Rien van de Weygaert (cosmology and large-scale structure formation)
Diederik Roest (string cosmology)
Aernout van Enter (statistical mechanics)
Thesis: Statistics of caustics in large-scale structure formation

2009 - 2012 **Bachelor** Physics (*cum laude*), van Swinderen Institute, University of Groningen
2009 - 2012 **Bachelor** Astronomy (*cum laude*), Kapteyn Institute, University of Groningen
2009 - 2012 **Bachelor** Mathematics (*cum laude*), Bernoulli Institute, University of Groningen
Committee: Rien van de Weygaert (cosmology and large-scale structure formation)
Elisabetta Pallante (quantum field theory)
Gert Vegter (computational geometry)
Thesis: Analysis of Betti numbers and persistence diagrams of 2D Gaussian random fields

July 2016 **Summer school** It from qubit summer school, Perimeter Institute, Canada (two weeks)
July 2015 **Summer school** Prospects in theoretical physics: new insights into quantum matter, Institute for Advanced Study, Princeton (one week)

July 2015 **Summer school** Princeton summer school on condensed matter physics (one week)
August 2011 **Summer school** Dealing with environmental heritage, Bath, United Kingdom (two weeks)
July 2010 **Summer school** University of Cambridge international summer school in science (two weeks)

Awards

April 2020 Canadian Association of Physicists' Division of Theoretical Physics (DTP) and Winnipeg Institute for Theoretical Physics (WITP) P.R. Wallace PhD Thesis Prize
Best PhD thesis in Theoretical Physics from a Canadian university (2020).

November 2014 De Zeeuw-Van Dishoeck award 2014
Best master thesis in Astronomy from a Dutch university (2014).

July 2014 GUF-100 prize 2014
Best student in the Faculty of Mathematics and Natural Sciences at the University of Groningen (2014).

November 2011 Silver medal in the university physics competition 2011
A university competition in which groups of three physics students all over the world solve a problem and write an article in 48 hours.

September 2010 Young Talent encouragement prize 2010 in Physics
Prize awarded by the Koninklijke Hollandse Maatschappij der Wetenschappen (Royal Holland Society of Sciences and Humanities) for the best freshman Physics student at the University of Groningen in 2009-2010.

November 2007 Third place in the CanSat competition
The CanSat project is an annual competition organized by the Delft University of Technology. Teams of secondary school students design and build a satellite in a Coca-Cola can.

Scholarships

March 2015 University of Waterloo scholarship, for PhD at the Perimeter Institute
July 2014 Hendrik Muller fund 2014: Scholarship for excelling Dutch students
June 2014 VSB fund 2014: Scholarship for Dutch students studying in abroad

Research Visits

November 2025 Research visit to Chris Howls and Ines Aniceto, Southampton University, Southampton, United Kingdom
October 2025 Research visit to Sandrine Codis, Université Paris-Saclay, Paris, France
September 2025 Research visit to Chris Howls and Ines Aniceto, Southampton University, Southampton, United Kingdom
May 2023 Research visit to Beatrice Bonga, Radboud University, Nijmegen, The Netherlands
March 2023 Research visit to Ue-Li Pen, Academia Sinica Institute of Astronomy and Astrophysics, Taipei, Taiwan
March 2023 Research visit to Beatrice Bonga, Radboud University, Nijmegen, The Netherlands
January 2023 Research visit to Beatrice Bonga, Radboud University, Nijmegen, The Netherlands
June 2022 Research visit to Rien van de Weygaert, Kapteyn Institute, Groningen, The Netherlands
January 2020 Research visit to Beatrice Bonga, Radboud University, Nijmegen, The Netherlands
June 2019 Research visit to Rien van de Weygaert, Kapteyn Institute, Groningen, The Netherlands
October 2018 Research visit to Rien van de Weygaert, Kapteyn Institute, Groningen, The Netherlands
May 2018 Research visit to Jean-Luc Lehners, Albert Einstein Institute, Potsdam, Germany
Feb - Apr 2017 Research visit to DAMTP, University of Cambridge, United Kingdom
March 2017 Research visit to Jean-Luc Lehners, Albert Einstein Institute, Potsdam, Germany

Publications & talks

Journal articles

2026 I. Aniceto, **J. Feldbrugge**, C. J. Howls "Which Saddles Contribute? The South-East Rule for Multidimensional Integrals", (2026) arXiv:2606.28271 [math-ph]

2026 A. Read, **J. Feldbrugge**, C. Boehm, R. van de Weygaert, B. Hertzsch "Caustic Skeleton and the Local Cosmic Web: the Coma Cluster node and the Pisces-Perseus ridge", (2026) arXiv:2604.22213 [astro-ph.CO]

2026 B. Hertzsch, **J. Feldbrugge**, R. van de Weygaert "Galaxy Populations in the IllustrisTNG Caustic Skeleton", (2026) *MNRAS* arXiv:2604.18209 [astro-ph.CO]

2026 **J. Feldbrugge,**, S. Crew, U-L Pen, "Resurgence and Hyperasymptotics in Wave Optics Astronomy", *JCAP* (2026), Issue 22 arXiv:2602.21493 [astro-ph]

2026 A. Weber, **J. Feldbrugge**, E. Pisanty, "A universal approach to saddle-point methods in attosecond science", Physical Review A (2025), arXiv:2510.12545 [quant-ph]

2025 B. Hertzsch, **J. Feldbrugge**, Maé Rodriguez and Rien van de Weygaert, "A New Recipe for Caustic Pancakes: On the Reality of Walls in the Cosmic Web", *JCAP* (2025), arXiv:2510.02419 [astro-ph.CO]

2025 **J. Feldbrugge** and U-L Pen, "The real-time Feynman path integral for step potentials", *Annals of Physics* (2025), arXiv:2508.17578 [quant-ph]

2025 **J. Feldbrugge** and J. Y. L. Jones, "Efficient evaluation of real-time path integrals", *Physical Review D* (2025) 111. arXiv:2501.16323 [quant-ph]

2025 B. Bonga, **J. Feldbrugge** and A. Metidieri, "Wave optics for rotating stars", *Physical Review D* (2025) Volume 111, Issue 6. arXiv:2410.03828 [astro-ph.CO]

2025 **J. Feldbrugge** and R. van de Weygaert, "What makes a cosmic filament? The dynamical origin and identity of filaments I. fundamentals in 2D", *Monthly Notices of the Royal Astronomical Society* (2025). arXiv:2405.20475 [astro-ph.CO]

2025 **J. Feldbrugge**, "Phase-Space Delaunay Tessellation Field Estimator", *Monthly Notices of the Royal Astronomical Society* (2025) no.1, 536. arXiv:2402.16234 [astro-ph.CO]

2023 **J. Feldbrugge**, D. L. Jow, U.-L. Pen, "Crossing singularities in the saddle point approximation", *Physical Review Letters* (2023, R&R). arXiv:2309.12427 [quant-ph]

2023 **J. Feldbrugge**, D. L. Jow, U.-L. Pen, "Complex classical paths in quantum reflections and tunneling", *Physical Review D* (2025) Volume 111, Issue 8, id.085027, 29 pp. arXiv:2309.12420 [quant-ph]

2023 **J. Feldbrugge**, N.M.D. Niezink, "Orthogonality relations for conical functions of imaginary order", (2023). arXiv:2309.05616 [math]

2023 **J. Feldbrugge**, "Complex evaluation of angular power spectra: Going beyond the Limber approximation", *Physical Review D* (2023) nr. 108, 103007. arXiv:2304.13064 [astro-ph.CO]

2023 **J. Feldbrugge**, Y. Yan, and R. van de Weygaert, "Statistics of tidal and deformation eigenvalue fields in the primordial Gaussian matter distribution: the two-dimensional case", *Monthly Notices of the Royal Astronomical Society* (2023). arXiv:2301.07200 [astro-ph.CO]

2023 **J. Feldbrugge**, and R. van de Weygaert, "Cosmic web & caustic skeleton: non-linear constrained realizations - 2D case studies", *Journal of Cosmology and Astroparticle Physics* (2013) no.2, 58. arXiv:2212.07840 [astro-ph.CO]

2023 **J. Feldbrugge**, U.-L. Pen, and N. Turok, "Oscillatory path integrals for radio astronomy," *Annals of Physics* (2023) no.451, 169255. arXiv:1909.04632 [astro-ph.HE]

2023 **J. Feldbrugge**, "Multi-plane lensing in wave optics," *Monthly Notices of the Royal Astronomical Society* (2023) nr.250, 2995-3006. arXiv:2010.03089 [astro-ph.CO]

2022 **J. Feldbrugge**, and N. Turok, "Existence of real time quantum path integrals", *Annals of Physics* (2023) arXiv:2207.12798 [hep-th]

2022 D. Jow, U.-L. Pen, and **J. Feldbrugge**, "Regimes in astrophysical lensing: refractive optics, diffractive optics, and the Fresnel scale", *Monthly Notices of the Royal Astronomical Society* (2022). arXiv:2204.12004 [astro-ph.CO]

2021 G. Wilding, K. Nevenzeel, R. van de Weygaert, G. Vegter, P. Pranav, B.J.T. Jones, K. Efstathiou, and **J. Feldbrugge**, "Persistent homology of the cosmic web. I: Hierarchical topology in ΛCDM cosmologies" *Monthly Notices of the Royal Astronomical Society* (2020) nr.507, 2968-2990. arXiv:2011.12851 [astro-ph.CO]

2020 **J. Feldbrugge**, and N. Turok, "Gravitational lensing of binary systems in wave optics," *Physical Review Letters* (2020, R&R). arXiv:2008.01154 [gr-qc]

2019 **J. Feldbrugge**, M. van Engelen, R. van de Weygaert, P. Pranav, and G. Vegter, "Stochastic homology of Gaussian vs. non-Gaussian random fields: Graphs towards Betti numbers and persistence diagrams, *Journal of Cosmology and Astroparticle Physics* (2019) no.9, 52–100. arXiv:1908.01619 [astro-ph.CO]

2019 A. Di Tucci, **J. Feldbrugge**, J.-L. Lehners, N. Turok, "Quantum incompleteness of inflation," *Physical Review D, 100* (2019) no.6, 63517. arXiv:1906.09007 [hep-th]

2019 P. Pranav, R. van de Weygaert, G. Vegter, B.J.T. Jones, R.J. Adler, **J. Feldbrugge**, C. Park, T. Buchert, and M. Kerber, "Topology and geometry of Gaussian random fields I: on Betti numbers, Euler characteristic, and Minkowski functionals" *Monthly Notices of the Royal Astronomical Society, 485* (2019) no.3, 4167–4208. arXiv:1812.07310 [astro-ph.CO]

2018 **J. Feldbrugge**, J.-L. Lehners, and N. Turok, "Inconsistencies of the new no-boundary proposal," *Universe, 4* (2018), no.10, 100–115. arXiv:1805.01609 [hep-th]

2018 **J. Feldbrugge**, R. van de Weygaert, J. Hidding, and J. Feldbrugge, "Caustic skeleton & cosmic web," *Journal of Cosmology and Astroparticle Physics* (2018) no.05, 27–81. arXiv:1703.09598 [astro-ph.CO]

2018 **J. Feldbrugge**, J. Lehners, and N. Turok, "No rescue for the no boundary proposal: Pointers to the future of quantum cosmology," *Physical Review D, 97* (2018), no.2, 23509 arXiv:1708.05104 [hep-th]

2017 **J. Feldbrugge**, J.-L. Lehners, and N. Turok, "No smooth beginning for spacetime," *Physical Review Letters, 119* (2017), no.17, 171301. arXiv:1705.00192 [hep-th]

2017 **J. Feldbrugge**, J.L. Lehners, and N. Turok, "Lorentzian quantum cosmology," *Physical Review D, 95* (2017), no.10, 103508. arXiv:1703.02076 [hep-th]

2016 **J. Feldbrugge**, J. Hidding, and R. van de Weygaert "Statistics of caustics in large-scale structure formation," *The Zeldovich Universe: Genesis and Growth of the Cosmic Web, Proceedings of the International Astronomical Union, IAU Symposium, 308* (2016), 107–114. arXiv:1412.5121 [astro-ph.CO]

2013 R. van de Weygaert, G. Vegter, H. Edelsbrunner, B.J.T. Jones, P. Pranav, C. Park, W. Hellwing, B. Eldering, N. Kruithof, E.G.P. Bos, J. Hidding, **J. Feldbrugge**, E. ten Have, M. van Engelen, M. Caroli, and M. Teillaud, "Alpha, Betti and the megaparsec universe: On the topology of the cosmic web," *Transactions on Computational Science XIV: Special Issue on Voronoi Diagrams and Delaunay Triangulation. Lecture Notes in Computer Science, Vol. 6970. Springer Berlin Heidelberg* (2013). arXiv:1306.3640 [astro-ph.CO]

Theses

2019 **J. Feldbrugge**, "Path integrals in the sky: classical and quantum problems with minimal assumptions," PhD thesis, Perimeter Institute, University of Waterloo, supervised by N. Turok. Available online.

2015 **J. Feldbrugge**, "Primordial non-Gaussianity and large-scale structure," Part III Essay, University of Cambridge, supervised by P. Shellard and T. Giannantonio. Available online.

2014 **J. Feldbrugge**, "Statistics of caustics in large-scale structure formation," Master thesis, University of Groningen, supervised by R. van de Weygaert, D. Roest, A.E. van Enter. Available online.

2012 **J. Feldbrugge** and M. van Engelen, "Analysis of Betti numbers and persistence diagrams of two-dimensional Gaussian random fields," Bachelor thesis, University of Groningen, supervised by R. van de Weygaert, E. Pallante, G. Vegter. Available online.

Invited talks

August 2026 Astronomy Department of The Physics Institute of the Federal University of Rio Grande do Sul
Seminar: Unravelling the cosmic web with caustics

July 2026 Nancy, France, EXCOSM workshop "Random Structures and Patterns in Spatial Data: Here, There, and Everywhere"
Presentation: The caustic skeleton of the cosmic web

May 2026 Physics and Astronomy, Nottingham, United Kingdom
Seminar: Real-Time Path Integrals, Caustics and Interference in Cosmology

May 2026 The Center of Gravity – Niels Bohr Institute, Copenhagen, Denmark
Seminar: Real-Time Path Integrals, Caustics and Interference in Cosmology

April 2026 Cosmological Frontiers in Fundamental Physics 2026, Perimeter Institute, Waterloo, Canada
Presentation: Real-Time Path Integrals, Caustics and Interference in Cosmology

December 2025 King's College London, London, United Kingdom
Presentation: Real-Time Path Integrals, Caustics and Interference in Cosmology

November 2025 University of Southampton, Southampton, United Kingdom
Seminar: Unravelling the cosmic web with caustics

November 2025 University of Vienna, Vienna, Austria
Presentation: Understanding the cosmic web with caustics

October 2025 Université Paris-Saclay, Paris, France
Presentation: Understanding the cosmic web with caustics

October 2025 University of Bonn, Bonn, Germany
Presentation: Real-Time Path Integrals, Caustics and Interference

October 2025 Université Paris-Saclay, Paris, France
Presentation: Real-Time Path Integrals, Caustics and Interference in Cosmology

September 2025 Cosmology seminar at Perimeter Institute for Theoretical Physics, Waterloo, Canada
Presentation: Real-Time Path Integrals, Caustics and Interference in Cosmology

August 2025 Seminar at the Canadian Institute for Theoretical Astrophysics, Toronto, Canada
Presentation: Understanding the cosmic web with caustics

June 2025 The Cosmic Web from Galaxies to Cosmology, Institute for Fundamental Physics of the Universe, Trieste, Italy
Presentation: Caustics and Interference

March 2025 Seminar at the University of Pennsylvania, Philadelphia, United States,
Presentation: Real-time Path Integrals, Caustics and Interference

October 2024 Complexity and Cosmos, Gran Sasso Science Institute, L'Aquila, Italy
Presentation: Path integrals in the sky

June 2024 Singular and Oscillatory Integrals, University College London
Presentation: Integration in the complex plane with Picard-Lefschetz theory

May 2024 Tuorla-Tartu meeting 2024: Borderless Universe
Presentation: What makes a cusp/filament?

December 2023 Gauge-Gravity by the ghats' seminar, Center for High Energy Physics (CHEP), Indian Institute of Science, India.
Presentation: Complex classical paths in quantum reflections and tunnelling

December 2023 Large-scale parity violation, Academia Sinica, Institute of Astronomy and Astrophysics, Taipei, Taiwan,
Presentation: Dissecting the cosmic web with caustics

October 2023 International Loop Quantum Gravity Seminar
Presentation: Complex saddle points in gravitational path integrals

September 2023 Theoretical Physics seminar Newcastle, University of Newcastle
Presentation: On the existence of real-time path integrals

September 2023 Complexity and Cosmos, Gran Sasso Science Institute, L'Aquila, Italy
Presentation: On the existence of real-time path integrals

July 2023 Quantum Gravity 2023, Radboud University, Nijmegen, The Netherlands
Presentation: On the existence of real-time path integrals

July 2023 Theoretical physics group, University of New Brunswick, New Brunswick, Canada
Presentation: On the existence of real-time path integrals

March 2023 Optimal Transport Theory and Applications to Physics, Ecole Physique, Les Houches, France
Presentation: Dissecting the cosmic web with caustics

September 2022 2nd Roman Juskiewicz Symposium, Nicolaus Copernicus Astronomical Center, Warsaw, Poland
Presentation: Dissecting the cosmic web with caustics

June 2022 Information Universe 4, University of Groningen, Groningen, The Netherlands
Presentation: Dissecting the cosmic web with caustics

May 2022 Cosmology seminar, Oxford University, Oxford, England
Presentation: Dissecting the cosmic web with caustics

May 2022 UK Cosmo Meeting 2022, Newcastle University, Newcastle, England
Presentation (keynote): Dissecting the cosmic web with caustics

October 2021 Higgs hour, University of Edinburgh, Edinburgh, Scotland
Presentation: Interference, caustics and oscillatory integrals

May 2021 Seminar Universidad Nacional Autonoma de Mexico, Mexico city
Presentation: Interference phenomena in lensing and quantum physics

May 2021 Sirius A symposium 2021: To infinity and beyond
Presentation: The caustic skeleton of the cosmic web

October 2020 Quantum & Gravity Seminar, Radboud Universiteit, Nijmegen, The Netherlands
Presentation: Lorentzian quantum cosmology

October 2020 Pusar group meeting, CITA, Toronto, Canada
Presentation: Multi-plane lensing and gravitational binary lensing in wave optics

November 2019 Scintillometry 2019, Max Planck Institute for Radio Astronomy, Bonn, Germany
Presentation: Oscillatory path integrals for radio astronomy

June 2018 2018 Rotman summer institute in philosophy of cosmology, Godrich, Canada
Presentation: Lorentzian quantum cosmology (two lectures)

September 2016 CITA-PI day: Gravitational non-linear instability, CITA, Toronto, Canada
Presentation: Shocks in the early universe and gravitational waves

Contributed talks

June 2025 Cosmology from home 2025
Presentation: What makes a cosmic wall/filament in the caustic skeleton of the cosmic web?

June 2025 Triangular Conference on Cosmological Frontiers in Fundamental Physics 2025, Paris, France
Poster: Real-Time Path Integrals

October 2024 Scintillometry 2024, University of Central Florida
Presentation: Lensing by rotating stars

July 2024 Relativistic effects and novel observables, University of Geneva
Presentation: What makes a filament/wall?

July 2024	Cosmology from home Presentation: What makes a filament/wall?
June 2024	Theoretical Modeling of the Large Scale Structure of the Universe, University of Edinburgh Presentation: What makes a filament/wall?
March 2024	Large Scale Structure group meeting Cambridge, University of Cambridge Presentation: Caustic skeleton: what makes a filament in the cosmic web?
February 2023	The Co-evolution of the Cosmic Web and Galaxies across Cosmic Time, Kavli Institute for Theoretical Physics, UC Santa Barbara, California, United States Presentation: Dissecting the cosmic web with caustics
July 2022	Cosmology from Home Presentation: Dissecting the cosmic web with caustics
March 2022	Cosmic Cartography 2022: Exploring the Cosmic Web and Large-Scale Structure, Kavli IPMU, Kashiwa, Japan Presentation: The caustic web and non-linear constrained Gaussian random fields
October 2021	Tuorla-Tartu meeting 2021, University of Turku, Turku, Finland Presentation: Caustic skeleton of the cosmic web
September 2020	Cosmology from home 2020, virtual conference on all aspects of cosmology. Presentation: The caustic skeleton of the cosmic web
March 2020	Topological statistics group meeting. Department of statistics at Carnegie Mellon University, Pittsburgh, United States Two presentations: Cosmology and topology I, and Cosmology and topology II
February 2020	The centre for the universe Waterloo Centre for Astrophysics day, Waterloo, Canada Presentation: Path integrals for radioastronomy and gravitational lensing II
February 2020	Cosmology group meeting, Perimeter Institute, Waterloo, Canada Presentation: The caustic skeleton of the cosmic web
January 2019	The cosmic web in the local universe, Lorentz center, Leiden, The Netherlands Presentation: The caustics skeleton of the cosmic web
October 2019	The Future of Astronomy, Waterloo Centre for Astrophysics, Waterloo, Canada Poster: Oscillatory path integrals for radio astronomy
September 2019	Theory group meeting, Carnegie Mellon University, Pittsburgh, United States Presentation: Interference and Picard-Lefschetz theory
September 2019	Simplicity III, Perimeter Institute, Waterloo, Canada Presentation: Fun with path integrals II
August 2019	Graduate student meeting, Perimeter Institute, Waterloo, Canada Presentation: Oscillatory integrals in the complex plane
June 2019	Probabilities in cosmology, University of Groningen, The Netherlands Presentation: Lorentzian beginnings of the universe
June 2019	The cosmic web: from galaxies to cosmology, Edinburgh, United Kingdom Presentation: Caustic skeleton of the cosmic web
May 2019	Cosmology group meeting Perimeter Institute, Waterloo, Canada Presentation: lenses and oscillatory integrals
May 2018	Albert Einstein Institute group meeting, Potsdam, Germany Presentation: Classical and weak trajectories
November 2017	Path integral of gravity, Perimeter Institute, Waterloo, Canada Presentation: Quantum incompleteness of inflation II
September 2017	Cosmology group meeting CITA, Canada Presentation: The instability of the no-boundary proposal
July 2017	Cosmic web day, University of Toronto, Toronto, Canada Presentation: The caustic skeleton of the cosmic web
May 2017	PI-day, Perimeter Institute, Waterloo, Canada Presentation: Lorentzian quantum cosmology
May 2017	Theory Canada 12, York University, Toronto, Canada Presentation: Lorentzian quantum cosmology
May 2017	Cosmology group meeting Perimeter Institute, Waterloo, Canada Presentation: Caustics in large-scale structure
May 2017	String cosmology group meeting Van Swinderen Institute, Groningen, Netherlands Presentation: Lorentzian quantum cosmology
April 2017	New Thoughts 3: About the universe and more, Ely, United Kingdom Presentation: Lorentzian quantum cosmology

April 2017 British gravity meeting 2017, University of Oxford, Oxford, United Kingdom
Presentation: Lorentzian quantum cosmology

June 2016 Cosmology group meeting CITA, Canada
Presentation: Statistics of caustics in large-scale structure

May 2016 Cosmology group meeting Perimeter Institute, Canada
Presentation: Statistics of caustics in large-scale structure

March 2016 Statistics of extrema of large-scale structure, Lorentz center, Leiden, The Netherlands
Presentation: Statistics of caustics in large-scale structure

June 2014 IAUS 308: The Zel'dovich universe, Tallinn, Estonia
Presentation: Statistics of caustics in large-scale structure formation

October 2012 Structure of the cosmic web, Leibniz institute astrophysics, Potsdam, Germany
Presentation: Analysis of Betti numbers and persistence diagrams in 2D GRFs

CONFERENCES AND WORKSHOPS (ATTENDED ONLY)

March 2026 Max Planck-IAS-NTU Center Kick-off Conference, Institute for Advanced Science, Princeton, United States

August 2025 Charting the Future Symposium: Big questions in particle physics, strong gravity, and cosmology over the next 25 years, Perimeter Institute, Waterloo, Canada

January 2023 New Directions in Theoretical Physics 4, University of Edinburgh, Edinburgh, Scotland

June 2022 Analogue Models of Gravity and Fluctuation-Induced phenomena, University of Edinburgh, Edinburgh, Scotland

June 2022 Online Workshop "Physics of the Early Universe"

January 2021 Cosmology 2021: the rise of field theory, University of Cambridge, Cambridge, The United Kingdom

October 2020 The information universe: What is the role of information in our Universe? University of Groningen, Groningen, The Netherlands

January 2020 First Dutch Mathematical Relativity Day, Radboud University Nijmegen, Nijmegen, The Netherlands

September 2019 Cosmological frontiers in fundamental physics 2019, Perimeter Institute, Waterloo, Canada

November 2018 Quantum universe, in celebration of Neil Turok's 60th birthday, Centro de Estudio Cientificos (CECs), Valdivia, Chile

August 2018 Cosmology and gravitational physics with lambda, Nordita, Stockholm, Sweden

June 2018 Scanning new horizons: Emergent space-time, black holes and quantum information, Van Swinderen Institute, Groningen, The Netherlands

January 2018 Gravity in the early universe, Princeton University, Princeton, United States

June 2017 Bounce scenarios in cosmology, Perimeter Institute, Waterloo, Canada

January 2017 Fundamentals of the universe, Van Swinderen Institute, Groningen, The Netherlands

January 2017 New directions in theoretical physics II, Higgs center for theoretical physics, Edinburgh, United Kingdom

October 2016 Midwest relativity meeting, Perimeter Institute, Waterloo, Canada

June 2016 Time in cosmology, Perimeter Institute, Canada

June 2016 Concepts and paradoxes in a quantum universe, Perimeter Institute, Canada

June 2016 Cosmological frontiers in fundamental physics 2016, Perimeter Institute, Canada

June 2015 Convergence, Perimeter Institute, Canada

April 2014 Quantum universe 4, University of Groningen, The Netherlands

March 2013 Quantum universe 3, University of Groningen, The Netherlands

Advising

2025 - present Advisor PhD project, Maé Rodriguez, University of Edinburgh, Edinburgh, United Kingdom: **Evaluating real time path integrals for relativistic systems**

2025 - present Advisor PhD project, Amelie Reader, University of Sydney, Sydney, Australia: **Probing cosmological redshift surveys with the caustic skeleton**

2023 - present Advisor PhD project, Benjamin Hertzsch, University of Edinburgh, Edinburgh, United Kingdom: **Probing cosmological redshift surveys with the caustic skeleton**

2023 - present Co-advisor PhD project, Anne Weber, University College London, London, United Kingdom: **Complex trajectories in laser ionization and recombination experiments**

2023 - 2025 Co-advisor PhD project, Johanna Borissova, Perimeter Institute, Waterloo, Canada: **Lorentzian worm holes**

2023 - 2025 Co-advisor PhD project, Ariadna Metidieri, Radboud University, Nijmegen, The Netherlands: **Lensing of rotating stars and photons in wave optics**

2022 - present Co-advisor PhD project, Joshua Jones, Dublin Institute for Advanced Studies, Dublin, Ireland: **Real-time**

path integrals

2024 Advisor master project, Maé Rodriguez, University of Edinburgh, Edinburgh, United Kingdom: **Hamiltonian Monte Carlo sampling of primordial caustic constraints**

2023 Advisor master project, Vera Li, University of Edinburgh, Edinburgh, United Kingdom: **The caustic skeleton and redshift space distortions**

2022 - 2023 Co-advisor bachelor project, Yonatan Sklansky, University of Pennsylvania, United States: **Topology of multi-stream regions in N-body simulations**

2022 - 2023 Advisor master project, Yihan Yan, University of Waterloo, Canada: **Statistics of critical points in eigenvalue fields**

2020 - 2022 Co-advisor PhD project, Georg Wilding, University of Groningen, The Netherlands: **Topology of the two-dimensional cosmic web**

2021 - 2022 Advisor bachelor project, Yihan Yan, Waterloo University, Canada: **Homology of two- and three-dimensional Gaussian random fields**

2020 - 2021 Co-advisor PhD project, Varun Rustagi, University of Groningen, The Netherlands: **The caustic skeleton of the local universe**

2018 - 2022 Co-supervisor master project Kevin Bixerman, University of Groningen, The Netherlands: **The caustic skeleton of the local universe**

Teaching

July 2025 EXCOSM Summer School, Large-scale strucutre of the Universe: from galaxies to cosmology, Haapsalu, Estonia

June 2023 School on Fundamentals of the Universe, Lorentz Center, Leiden, The Netherlands

June 2018 2018 Rotman summer institute in philosophy of cosmology, Godrich, Canada
Two lectures on Lorentzian quantum cosmology

January 2018 PSI winter school, co-supervisor project 'Pair creation in de Sitter spacetime', Huntsville, Canada

2010 - 2015 Physics lecturer, exam training for secondary school students, UOCG Market BV

2013 - 2014 Developer of teaching material for exam training in physics, UOCG Market BV

2008 - 2014 Tutoring in mathematics and physics for secondary school students

Service

2022 Co-organizer of the conference New Directions in Theoretical Physics 4 at the University of Edinburgh

2018 - present Reviewer for Physical Review Letters (PRL), Physical Review D (PRD), the Journal of Cosmology and Astroparticle Physics (JCAP), the Journal of High Energy Physics (JHEP), Monthly Notices of the Royal Astronomical Society (MNRAS), and Universe.

2013 - 2014 Chairman of the professor Hendrik de Waard foundation

2012 - 2013 Treasurer of the professor Hendrik de Waard foundation

2011 - 2012 Member of the education committee mathematics, chairman of the student council

2010 - 2014 Guide at the Gratama telescope of the Blaauw observatory, Groningen, at stargazing events

Languages

Dutch	Mother tongue	English	Fluent
German	Elementary	French	Elementary
Latin	Elementary		

Programming experience

Mathematica	Fluent	C++	Fluent
Python	Fluent	Matlab	Moderate
Swift	Moderate	Julia	Fluent
Fortran	Moderate	R	Elementary